Theory of Computation

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Let us discuss each one by one-

i) Negation

This proposition is also called NOT proposition. It is denoted by I or ~ or having horizontal bar on letter.

If S is a proposition then S defines NOT S.The truth table for this connective is

S	1s
T	F
F	T

ii) Conjunction (AND)

If A and B are two propositions then conjunction of A and B is denoted by A^B. The truth table for this connective is

A	B	A^B
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction OR

If A and B are two propositions then disjunction of A and B is denoted by AvB. The truth table for this connective is

A	B	AvB
T	T	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition. It can be denoted

byA-BorA=B.

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The truth table for this connective is

A	B	A-B
T	T	T
	F	F
T		
F	T	T
F	F	T

For example: Find truth value for folowing proposition.

If an apple is not green then it must be red.

Let, A denotes "An apple is not green" and B denotes It must be red" The statement

A is false and B is true. Hence the proposition is true.

v) If and only if

If A and B are two propositions then A if and only if B.It can be denoted as AB

The truth table will be

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	T

For exampleI will go to the school if and only if it is not raining.Here statement A

is I will go to the school. The statement B is It is not raining Thus the resultant

proposition is false.

1.5 Relations

Relaonship is a major aspect between two objects, even this is true in our real life

One ohfect can be related with the other object by a some relation. Then those two objects

forma pair based on this certain relationship.

Definition The relation R is a collction for the set S which represents the pair of

elements.

For example: (a, b) is in R. We can represent their relation as a R b. The first

component of each pair is chosen from a set called domain and second component of each

pair is chosen from a set called range.

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Properties of Relations

A relation R on set S is

1.Reflexive if iRi for all i in S.

2.Irreflexive if iRi is false for all i in S.

3.Transitive if iRj and jRk imply iRk

4.Symmetric if iRj implies jRi.

5.Asymmetric if iRj implies that jRi is false.

Every asymmetric relation must be irreflexive

For example: if A=ab| then

reflexive relation R can be=a.a)b, b

irreflexive relation R can be=|a.b)1

transitive relation R can be=a, b).b, a)a.a1

symmetric relation R can be=ab).b, a

asymmetric relation R can be =a, b

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive

over some set S.

Suppose R is a set of relations and S is the set of elements.

For example: S is the set of lines in a plane and R is the relation of lines intersecting

to each other.

A relation is said to be antisymmetric if a, b)e R and b, ae R then a=

Example 1.5: Determine whether R is equivalence relation or not where

A=0121R=001011).2, 2)2, 1

Solution: The R is reflexive because 0, 0.1, 1)2.2E R.

Where as R is not symmetric because (0, 1) is not in R whereas (1.0) is in R.

Hence the R is not a equivalence relation.

Closures of Relations

Sometimes when the relation R is given it may not be reflexive or transitive.

By adding some pairs we make the relation as reflexive or transitive.

For exampleLetab)bcaabb1